

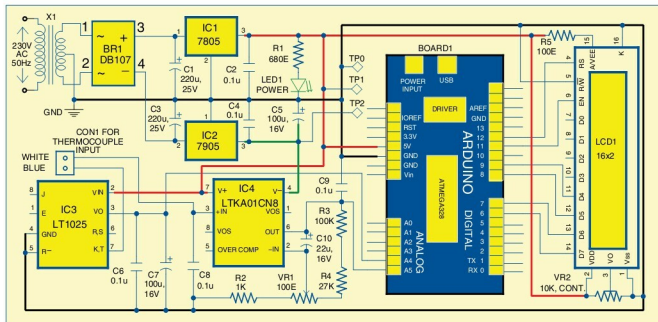


Fig. 3: Circuit diagram for interfacing a thermocouple with Arduino

TABLE I
Typical Examples of Selected Popular Thermocouples

Thermocouple type	Positive conductor	Negative conductor	Temperature range (°C)	Seebeck coefficient at +20 °C
J	Chromel	Constantan	0 to 760	51µV/°C
K	Chromel	Alumel	-200 to +1370	40.6µV/°C
E	Chromel	Constantan	-100 to +1000	62µV/°C
S	Platinum (10 per cent Rhodium)	Rhodium	0 to 1750	7µV/°C

phers. These thermocouples have a low Seebeck coefficient and relatively high cost. S type thermocouples allow measurements with precision up to $\pm 1^\circ\text{C}$ and require a relatively complex linearisation calculation algorithm.

Output of K type thermocouple is shown in Fig. 2. Slight non-linearity is reflected in the output due to properties of metal at increased temperature.

Circuit and working

Circuit diagram of the circuit for interfacing a thermocouple with Arduino is shown in Fig. 3. It is built around 230V AC primary to 12V, 300mA secondary transformer X1, bridge rectifier DB107 (BR1), +5V voltage regulator 7805 (IC1), -5V voltage regulator 7905 (IC2), cold-junction compensator

LT1025 (IC3), op-amp LTKA01CN8 (IC4), Arduino UNO (Board1) and a few other components.

230V AC mains is stepped down to 12V AC, 300mA through step-down transformer X1. Secondary output of X1 goes to bridge rectifier BR1. Capacitors C1 and C3 filter the ripples and the rectified output is given to regulators 7805 (IC1) and 7905 (IC2). IC1 provides +5V and IC2 provides -5V regulated DC output to operate the circuit. LED1 is used as a power-on indicator.

IC3 gives K type output at its pin 7. Operating temperature range of IC3 is -55°C to 125°C . Vo pin 3 of IC3 is externally pulled down with two capacitors so that it can deliver proper outputs for temperature below 0° . Vo signal is fed to the analogue pin (A4) of Board1.

PARTS LIST

Semiconductors:

- | | |
|--------|------------------------------------|
| IC1 | - 7805, 5V regulator |
| IC2 | - 7905, -5V regulator |
| IC3 | - LT1025 cold-junction compensator |
| IC4 | - LTKA01CN8 op-amp |
| BR1 | - DB107 bridge rectifier |
| BOARD1 | - Arduino UNO (ATmega328) |
| LED1 | - 5 mm LED |
| LCD1 | - 16×2 alphanumeric display |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|-----|----------------------|
| R1 | - 680-ohm |
| R2 | - 1-kilo-ohm |
| R3 | - 100-kilo-ohm |
| R4 | - 27-kilo-ohm |
| R5 | - 100-ohm |
| VR1 | - 100-ohm potmeter |
| VR2 | - 10-kilo-ohm preset |

Cenozoic:

- C1, C3 - 220 μ F, 25V electrolytic
C2, C4, C6, C8, C9 - 0.1 μ F ceramic disk
C5, C7 - 100 μ F, 16V electrolytic
C10 - 22 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector terminal
- X1 - 230V AC primary to 12V, 300mA secondary transformer
- K type thermocouple

The thermocouple is connected to connector CON1. This is the input point for IC3 and IC4. Input could be from any thermocouple. This project was tested with J type and K type thermocouples.

Net output is amplified by IC4 for 5mV/°C sensitivity. For higher sen-